

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | Kent, OH

PROFESSIONAL SUMMARY

Full-stack AI engineer (MS CS, 2.5+ years) with Python, React/TypeScript, SQL, API integration, and LLM/prompt engineering experience. Delivered production AI systems with measurable reliability gains (~35% retrieval precision improvement, ~30% fewer post-deploy defects). Comfortable translating messy, inconsistent real-world data into reliable AI pipelines; experienced with customer stakeholder communication at technical and executive levels. Open to Munich relocation with EU work authorization support.

CORE COMPETENCIES

Full-Stack Engineering (Python, React, TypeScript, SQL) • LLM & Prompt Engineering • REST API Integration • Data Wrangling & Pipeline Engineering • AI Workflow Configuration • Customer Implementation Delivery • Rapid Prototyping • FastAPI / Docker

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting

- Configured and deployed **AI-enabled LLM workflows** using prompt engineering and domain-specific tuning; improved stability ~25%, reduced hallucinations >30%.
- Built data preprocessing pipelines for **messy EHR extracts** and support ticket logs (Pandas, NumPy); raised dataset reliability above **98%**.
- Built RAG systems integrating structured and unstructured enterprise data; ~35% **precision improvement**.
- Packaged inference as **FastAPI/Flask RESTful APIs** with Docker; integrated with enterprise systems via REST; reduced post-deploy defects ~30%.
- Daily stakeholder engagement across technical and non-technical levels in healthcare enterprise context.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- Optimized SQL stored procedures powering enterprise Order-to-Cash (ERP analog); ~20% latency reduction.
- Jenkins CI/CD; Grafana dashboards; RBAC and audit logging for compliance-sensitive environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

- Flask-based API deployment; transformer ML; RAG document Q&A for non-technical stakeholders.

KEY PROJECTS

E-Farming Digital Marketplace - Full-Stack Founder

PHP/MySQL/Bootstrap AgriTech marketplace independently designed and shipped. Cart, reviews, community blogging. **80-120 active users** onboarded in phase one. Demonstrates full-stack product ownership, customer implementation, and end-to-end delivery.

Dream Decoder - Full-Stack AI Application

FastAPI backend + React/TypeScript/Vite/Tailwind frontend. API orchestration (Perplexity Sonar + Replicate). Intermediate prompt transformation layers: ~30% **contextual alignment improvement**. Demonstrates full Python + React + API integration stack.

Patient Records Management - Healthcare Data Application

Bottle + SQLite relational CRUD app with normalized schema, foreign-key constraints, and aggregate SQL queries. Healthcare data entry and retrieval - enterprise data structure analog.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, React, SQL, JavaScript, PHP

LLM & AI: Prompt engineering, RAG, LangChain, LlamaIndex, vector search, agentic workflows

APIs & Infra: FastAPI, Flask, Docker, REST APIs, Jenkins, Grafana

ML: PyTorch, scikit-learn, XGBoost, HuggingFace

Work Authorization: F-1 OPT (US-authorized); open to EU work authorization / Munich relocation support